

Application of Advanced Frameworks for Data Imputation in Conflict Databases

A comparison of the bias introduced by MICE, Dataset Fusion and Listwise Deletion

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This paper evaluates the application of Multivariate Imputation by Chained Equations (MICE) to address the pervasive issue of missing data within the Global Terrorism Database (GTD). Critical variables, such as the number of perpetrators, exhibit high rates of missingness that are Missing Not At Random (MNAR) due to the clandestine nature of non-state actors and systematic reporting biases. Consequently, this research critically measures the accuracy, predictive validity, and inherent biases introduced by MICE when synthesizing categorical perpetrator data, specifically distinguishing between independent single actors, small cells, and large groups. By evaluating the imputed dataset’s performance metrics against available baseline records and utilizing Zero-Inflated Negative Binomial (ZINB) modeling to account for casualty overdispersion, this study quantifies the confounding effects of imputing MNAR fields. The findings expose the limitations and statistical biases inherent in algorithmic data recovery strategies, ultimately establishing a rigorous framework for evaluating the validity of data synthesis in longitudinal counterterrorism research.

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1 Introduction

The term “lone wolf” was popularized in the late 1990s by white supremacists Tom Metzger and Alex Curtis as part of an encouragement to fellow racists to act alone in committing violent crimes for tactical reasons (Bakker and Graaf 2010), fomenting a pivotal shift in the tactics employed of terrorism. From 2003 to 2018, the proportion of individual lone wolf attackers increased from 35.4% to 85%. Casualties inflicted by lone wolves in 2025 accounted for 93% of the total casualties recorded in Western Countries, demonstrating a continuing trend. (Institute for Economics & Peace 2025)

There is a critical need to examine the differences in tactics and outcomes of lone wolves, however the existing data is not immediately conducive to a quantitative analysis due to 83% of recorded attacks not accurately reporting on the number of perpetrators, making a differentiation of attacks based off the number of perpetrators difficult. Analysis is further confounded by the data being Missing Not At Random (MNAR).

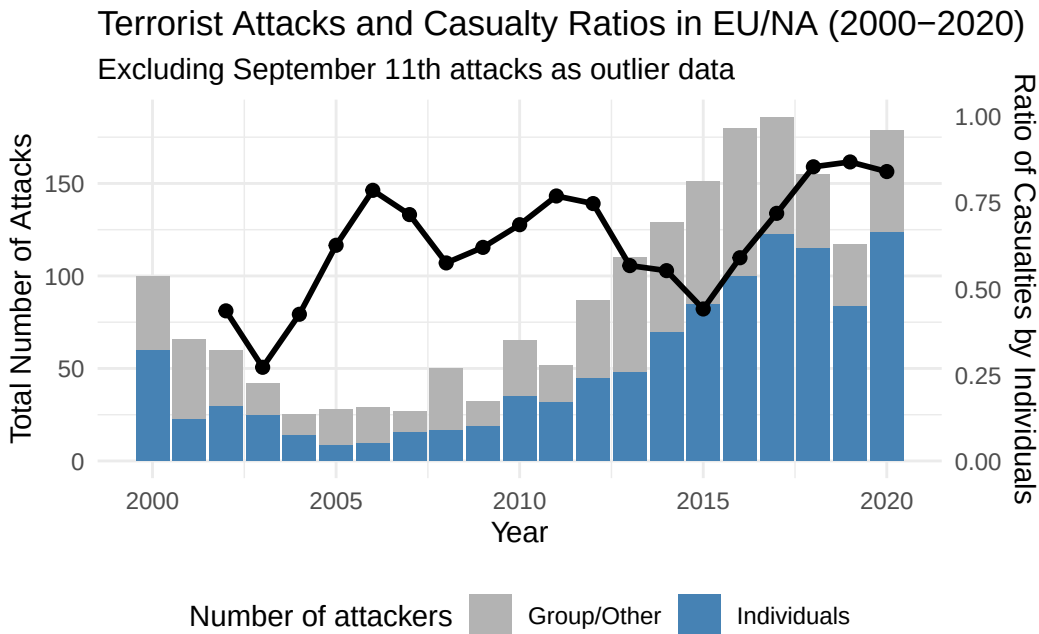
My research evaluates the accuracy of Multiple Imputation by Chained Equations (MICE), a powerful technique that uses tailored prediction methods for every missing column per record. The outcomes of the results of MICE imputation are measured against dataset fusion and listwise deletion approaches, common methods used to improve the accuracy of predictive models. The differences between results demonstrates the implicit bias introduced through imputation and synthesis methods on data that is not missing at random.

I sought to explore the difference in lethality, tactics, and outcomes of individuals performing attacks by themselves as compared to attacks perpetuated by multiple attackers. In producing a lens through which the impact and profile of “Lone Wolves” can be approached by counterterrorism researchers, methods of identification and risk assessment can be performed. I then examined the

changes in individual-sized terrorist attacks over time, looking for developing trends in attack types, success rates, and casualties, seeking to explore the rate at which these attacks are increasing in their risk.

Finally, I compare my results against the imputed dataset with the results from the limited data available in the GTD on individual attacks. While the scarcity of this data has historically presented difficulties in analysis, the value in comparing the difference in predictors from the imputed dataset reveals the confounders introduced by MICE, belying the high accuracy results of MICE’s predictive model. Throughout this process I seek to ensure the ability for others to reproduce my results, as well as evaluate their validity by applying the Shapley Additive Explanation (SHAP) process to document the weights and reveal additional insight about lone wolf terrorism in comparison to other attack types.

Figure 1: Attacks and casualties by year in Western Countries, 2000-2020



2 Prior Works

The compound societal and economic issues inflicted upon a community by a terrorist attack are profound, leaving lasting scars that affect the stability of a region. Predicting terrorist attacks and establishing a model that connects terrorist actors to outcomes is a much-researched problem with often no clear solution (Wani 2024). Recent breakthroughs in statistical learning methods have opened new pathways into modeling existing data sources, including more accurate data synthesis mechanism, new opportunities for the fusion and augmentation of data from external sources, and the use of contextual models to refine and establish relationships between data.

Because terrorist attacks are inherently clandestine events often involving deliberate deception to evade law enforcement—particularly in the case of independent “lone wolf” actors—critical details regarding perpetrators, weapons, and exact casualties are frequently obscured or systematically underreported. Consequently, this lack of data is MNAR because the “missingness” is inextricably linked to the nature of the attack itself; the very characteristics that make a covert, low-profile attack successful directly cause the absence of detailed public reporting and the resulting informational voids within the database. This behavior is exemplified in the Global Terrorism Database (GTD), the primary source for my research data. (National Consortium for the Study of Terrorism and Responses to Terrorism (START) 2021)

2.1 About the Global Terrorism Database

The systematic quantification of political violence relies upon the structural integrity of longitudinal archives, among which the Global Terrorism Database (GTD) serves as the preeminent unclassified repository for researchers, policymakers, and security analysts (Abdalsalam et al. 2024b).

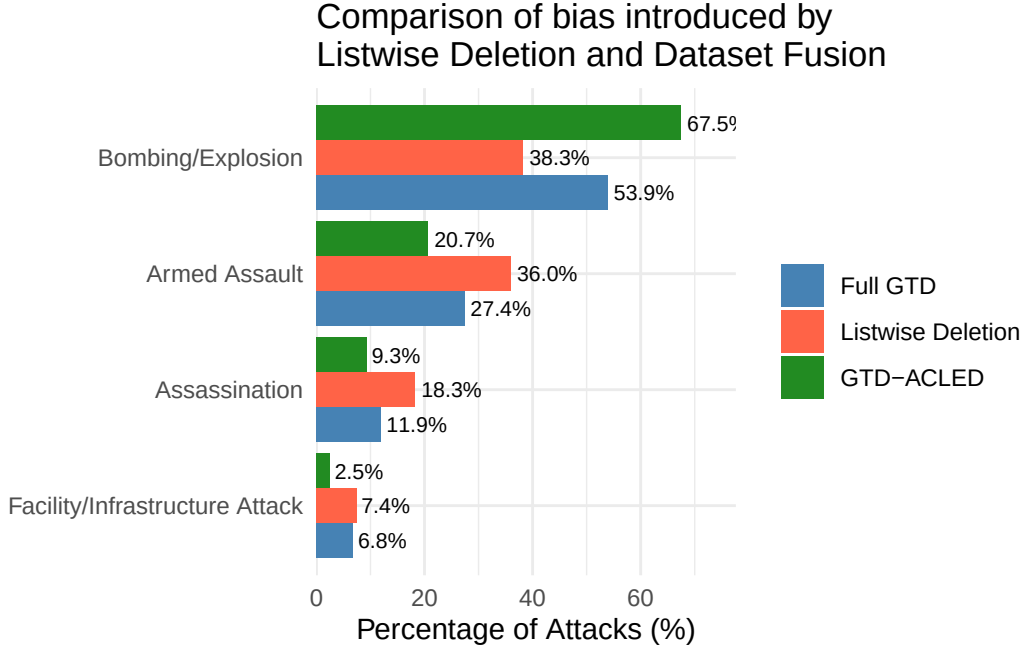
Maintained by the National Consortium for the Study of Terrorism and Responses to Terrorism

(START), the GTD documents over 180,000 incidents since 1970, capturing more than 130 variables for contemporary events, ranging from tactical specifics to perpetrator identities and casualty metrics (National Consortium for the Study of Terrorism and Responses to Terrorism (START) 2021). The GTD has been leveraged extensively to produce insightful analysis on the relationship between terrorism, political violence, and civil unrest, with LaFree and Dugan writing several seminal works that show terrorists as rational decision makers who are affected by counterterrorism controls. Despite its breadth of coverage and the wide array of descriptive fields to perform analysis upon, the utility of the GTD is persistently challenged by the “missingness” problem: the presence of null or unknown entries in critical fields such as perpetrator group names, casualty counts, and specific weapon sub-types. These informational voids are not merely random artifacts of data entry but are often the product of systematic biases in media reporting, state-sponsored censorship, or the inherent clandestine nature of non-state actors (Wani 2024). The GTD codebook takes a measured approach with the inclusion of a terrorist attack, choosing a pathway of preferring inclusion of an attack so long as it is not clearly disqualified, even if the record would have a relatively sparse level of populated fields.

The modern frontier of data science offers a robust suite of solutions to mitigate information gaps through advanced data imputation (Azad, Bosnić, and Kukar 2025). Traditional methods, such as listwise deletion or mean substitution, are increasingly viewed as inadequate because they frequently introduce significant bias and reduce the statistical power of subsequent analyses (Aleryani, Wang, and Iglesia 2020).

In the case of listwise deletion, the MNAR nature of the data results in an uneven distribution of missing results by attack and target type. For instance, the rate of missingness is lower for armed attacks against individuals as compared to bombings, where the number of perpetrators is far less likely to be known.

Figure 2: Distribution of Attack Types in the Full GTD (2000–2022)



Instead, a sophisticated integration of domain-specific Natural Language Processing (NLP) and meta-learning ensemble architectures is emerging as the gold standard for data recovery in conflict event databases. By leveraging extractive transformer models like ConflBERT to analyze unstructured event summaries and employing Meta-Imputation Balanced (MIB) frameworks to synthesize predictions from multiple algorithmic sources, researchers have extracted contextual information by collating common attacks with synthesized data to produce models capable of highly accurately predicting group name (Azad, Bosnić, and Kukar 2025). However, these papers and other quantitative analysis papers more interested in exploring the technical implementation and maximum efficacy of their imputation process often omit all but the largest terrorist groups, to produce groups of results that can be broadly analyzed using these contextual models (Pan 2021), focusing on groups with over 1000 documented attacks. In the case of one of Abdalsalam’s papers documenting the use of the BiGRU contextual model to identify terrorist groups, Abdalsalam’s dataset consisted of only terrorist groups with 500 or more documented attacks, and he also

employed the SMOTE-T data resampling technique to counter minority bias from underrepresented groups within this dataset of only 36 terrorist groups, demonstrating the confounding effect of missingness of the GTD’s dataset on some analysis and imputation techniques (Abdalsalam et al. 2024b).

Wani explores the mechanisms of missing data completely at random (MCAR), when the observed data is unevenly missing at random (MAR) and not missing at random (NMAR), in which the missingness is a result of a confounding effect (Wani 2024). In their review of potential imputation frameworks, Wani evaluates recovery strategies such as Multiple Imputation by Chained Equations (MICE), Random Forests, and Generative Adversarial Networks. Wani advocates for task-specific evaluation metrics rather than relying solely on standard error calculations to address imbalanced NMAR datasets such as the GTD which exhibit high dimensionality. In datasets such as these, fields are left blank or intentionally flagged as unknown if there is not clear consensus, even if some potential information points toward a value. Indeed, Abdasalam notes in their 2024 article exploring the missingness of the group name field that a value being left unknown is sometimes a conscious choice, stating “This gap in research may stem from the complex and dynamic nature of terrorist groups, which frequently evolve, change names, merge, or disband, thereby complicating the creation of accurate and current classification and prediction models.” (Abdalsalam et al. 2024b).

Reductions in the analyzed dataset have also prevailed to produce statistically significant predictions at the cost of external validity. Abdalsalam has had success with highly predictive contextual prediction models such as DistilBERT (Abdalsalam et al. 2021) or extractive text classifiers such as word2vec (Abdalsalam et al. 2024a), resulting in models capable of identifying attack type or location with a high degree of accuracy, although the research did exclude about 73% of the dataset to missing or insufficiently long descriptions. This exposes the limitations of

predicting a ‘freeform’ field such as group name using a statistical model without a limited Retrieval Augmented Generation (RAG) ability to narrow the search space of the contextual model. In the tangential field of Biomedical data, Azad has shown the promise of ensemble-based methods that combine contextual models with ensemble models to produced balanced approaches that adjust weights of predictors across multiple missing fields (Azad, Bosnić, and Kukar 2025). This approach may allow analysts to bridge the gap between records where generating a RAG of applicable context (such as English language news articles for models that are training against English language data) is not available.

Accounting for the level of missingness using a mask indicator is also useful to identify patterns in NMAR datasets. Using this method, Perez-Lebel benchmarked imputation approaches on healthcare databases and found that tree-based models utilizing Missing Incorporated in Attributes (MIA) provide the best balance of accuracy and speed (Perez-Lebel et al. 2022). When data is weakly correlated, a Ranked k-Nearest Neighbors (RkNN) can perform well. Khan, Shah, and Asim (2024) introduces the RkNN algorithm to refine traditional kNN imputation. By utilizing Ranked Set Sampling, this algorithm better handles data variability and reduces error when variables are weakly correlated or lack sufficient nearby neighbors.

Extensive research has been performed leveraging ensemble frameworks to reduce variance and improve prediction accuracy. Aleryani, Wang, and Iglesia (2020) proposed Multiple Imputation Ensembles (MIE), a system where multiple imputation techniques and ensemble methods collaborate to produce a superior final prediction when data is Missing Completely At Random. Azad, Bosnić, and Kukar (2025) detailed the Meta-Imputation Balanced (MIB) framework, which trains a supervised meta-model on synthetically masked data to intelligently weigh the outputs of multiple base imputers. These methods are automated, powerful tools, which trade computational performance for accuracy. A key advantage is the ability for the models to seek and identify

statistically significant fields on their own, although this can also be a hazard when colinear variables or keys are included for contention, reducing the external validity of the model to impute new data.

The authors above have all successfully used ensemble-based models of various flavors to produce highly accurate predictive models. However, maintaining external validity and reproducibility of the model is also critical. Štrumbelj and Kononenko discuss the process of evaluating the impact of ensemble models in the final product by adapting the Shapley value. The authors quantify the “fair” contribution of each feature value to a specific classification outcome, effectively revealing how much each variable influences a model’s decision. To resolve the exponential time complexity typically associated with these calculations, the paper proposes a sampling-based approximation method that makes the technique computationally feasible for complex datasets (Štrumbelj and Kononenko 2010). New compound fields can be generated against the GTD by analyzing text fields alongside restricting datasets to natural groups. Bridgelall describes an empirical text mining algorithm to produce a new “motive” field that allows analysts to categorize attacks based on underlying sentiments such as anti-government “rule,” race, or religion. The study employs advanced techniques like k-means co-clustering and nonlinear projections to visualize how these motives statistically intersect with specific tactics and targets (Bridgelall 2022). This can allow the conversion of freeform and numerical fields into factorized categories, allowing for additional analysis pathways.

Another option for field synthesis is the use of additional data sources connected without the direct tie between two fields. Cai et al. (2025) developed a sophisticated AI-driven tool called FAME (Fingerprint to Article Matching for Events) to systematically track how global media outlets report on natural disasters and terrorist attacks. By utilizing a “fingerprint” constructed from a specific combination of event class, location, and date, the system filters millions of

multilingual articles to identify those covering specific crises without the need for manual training data. This methodology reveals that news coverage is heavily biased toward wealthier nations and those with significant trade ties, often requiring much higher death tolls in poorer regions to receive equivalent attention. Ultimately, this work provides an open-source framework for computational social science to measure and understand the systemic factors that shape the global news agenda.

2.2 Hypothesis

The result of this robust analysis into the methods of accurately synthesizing data opens an opportunity to impute the number of perpetrators for a large slice of the GTD, omitting only the largest number of attack records to curb outliers. In predicting the number of perpetrators for every remaining record, an analysis of attacks predicted to be perpetrated by individuals will both expose recent trends on attack patterns and tactics, but also allow for a broader analysis of terrorism trends in regions such as North America and Europe which currently have a comparatively small number of records to quantitatively predict from. It is my prediction that the analysis of imputed data in comparison to the original dataset will reveal the confounding effect of the NMAR number of perpetrators to indicate which attack and weapon types have the highest degrees of casualty-inducing outcomes.

3 Process and Results

High-quality, varied data is the often-sought, rarely-acquired state for quantitative analysis. To limit the scope of my longitudinal analysis to roughly the present day conditions, I reduced the records of the GTD dataset to only records from 2000 to 2020, and I also excluded records with 50 or more perpetrators, as these records heavily skew the data. After the dataset had been scoped, I

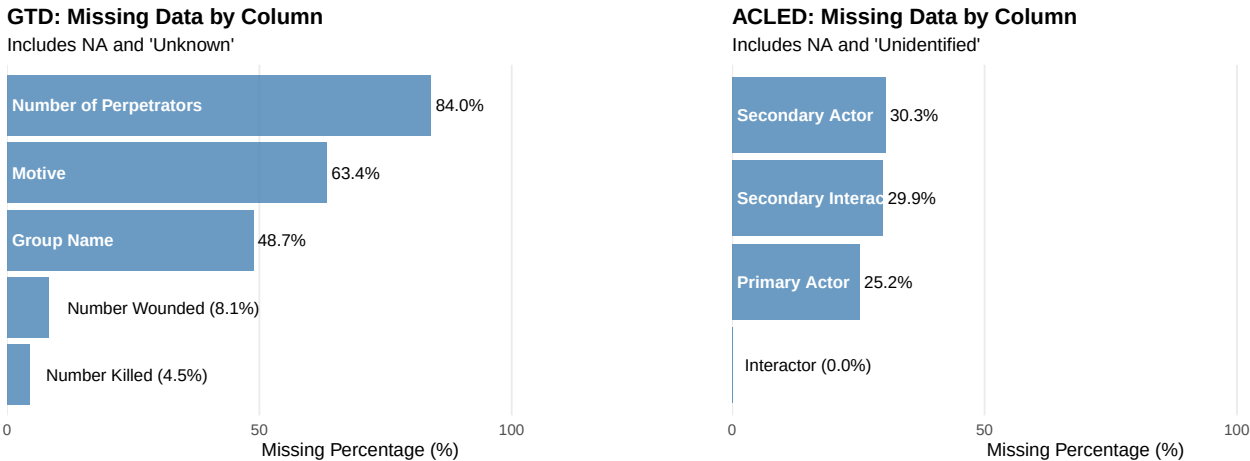
examined the remaining 138,000 records. Of these, 22,065 records (15.9%) had a defined number of perpetrators value, with 8,520 (5.9%) records having a single attacker. These rows comprise my test and training dataset.

The high degree of missing fields reflects a strategy by the GTD researchers at START to only include data they can establish with absolute certainty (LaFree and Dugan 2007). As a result, while information on attackers and outcomes might be fairly robust, if the accounts of an attack only refer to “a handful of attackers” or “an unknown assailant,” it is likely that the number of perpetrators would be left blank. I hypothesize that for the purposes of statistical analysis of outcomes, defining the exact number of actors is likely required; rather, establishing broader categories of Individual, Small Group and Large Group would be sufficient to convert the number of perpetrators into a categorical factor which could be used for imputation and prediction. In this way, the accuracy and specificity of the imputation model is greatly increased, without compromising the field for my purposes. By using Multivariate imputation by chained equations (MICE), I applied this method on the GTD dataset, with a few caveats beforehand: I assumed that attacks marked as suicide attacks in the GTD involved at least one perpetrator: The number of perpetrators must at least be nonzero in this case. I also restricted the variables used in my imputation model to primary attack type (`attacktype1`), primary weapon type (`weaptype1`), primary target (`targtype1`), and the broader region and country variables. I did not restrict my model to a specific time or country bounds, in the hopes that the resulting model would retain its validity independent of longitudinal or location confounding effects.

Examining the GTD data and existing research, I identified that the GTD data spanning from 2000-2020 has a number of critical fields such as Number of perpetrators, Group name and casualty information that have a high level of missingness. These fields are my primary candidates for imputation, and group name in particular has value in identifying not only groups involved, but

also in establishing which attacks currently defined as “unknown” could instead be attributed to independent “lone wolf” attackers, the propensity of which is increasing over time and represents a significant threat for counterterrorism law enforcement (Bakker and Graaf 2010).

Figure 3: Missing Data Percentages across GTD and ACLED Datasets



In an opportunity to provide additional context to the GTD, other adjacent datasets on terrorism can be compared. The Armed Conflict Location and Event Data (ACLED) database provides valuable context to the GTD when collated by location and time. The ACLED includes details on news sources describing the attribution (Armed Conflict Location & Event Data Project (ACLED) 2025) information as well as detailed actor information. While the ACLED contains nearly significantly more entries for the same time period, it is important to note the difference in membership criterion. The GTD contains information on confirmed non-state terrorist attacks where money was not the primary goal of the attack (LaFree and Dugan 2007); the ACLED has a much broader selection criterion including state actors and independent militias, as well as protest and riots that would be excluded from the GTD. A joined database of shared records brings the intersecting rows down to 20,782 records, removing 118,850 records from the potential comparison pool. In an effort to explore the confounding effect of the omission of the records not present in both datasets, my imputation workflow will be run on the standalone GTD dataset as well as a

joined dataset, giving opportunities for accuracy comparisons between the two, holding the workflow process consistent.

My research indicated that Multivariate imputation by chained equations (MICE) was the right combination of performance and accuracy to reliably perform data imputation on the number of perpetrators field, but the efficacy of the model greatly improves if the range of predicted options is more narrow than a numerical value. As a result, I categorized the number of perpetrators into 3 groups: individuals, small groups (2-5 actors), and large groups (6+). I then used the resulting model from the MICE process to impute the categorical factor of perpetrators for all records in my dataset.

3.1 Accuracy of MICE imputation against the GTD

After training, I generated a confusion matrix against the unimputed data for comparison purposes. The results of the MICE model reflect a relatively high degree of accuracy in predicting if an attack was perpetrated by an individual (97.7%), with decreasing performance for the small and large group categories. I then applied logistic regression against my imputed results to examine the differences in outcomes and tactics, if any, between attacks by individuals and groups. I determined that while individuals and groups have similar outcomes when the same attack and weapon types are considered, individual terrorist attackers are far more likely to employ vehicles and to commit suicide attacks. Region and Target Type were not strong predictors on their own, although comparisons within the region category show an increased probability of single actor terrorism in the Levant in comparison to Asia, where terrorist attacks involve more actors working in groups. Radiological weapons were excluded from the logistic regression visualization due to quasi-complete separation caused by insufficient event frequency.

Table 1: Performance Metrics: Predicting from Continuous Data vs Binned Categories

	Continuous Data		Binned Categories	
	Sensitivity	Specificity	Sensitivity	Specificity
Lone Wolf	0.391	0.879	0.699	0.730
Small Cell	0.475	0.588	0.468	0.702
Large Group	0.623	0.738	0.402	0.866

Performance Metrics of MICE Imputation by Category

A Kruskal-Wallis rank sum test was conducted to determine if there were differences in the number of casualties (nkill) based on the size of the perpetrator group. The results indicated a highly significant difference in the distribution of casualties between Lone Wolf, Small Cell, and Large Group attacks ($H(2) = 1210.5, p < 0.001$). To determine where the significant differences in casualties lay among the perpetrator categories, a post-hoc pairwise Wilcoxon rank sum test was conducted using the Holm method to control for multiple comparisons. The analysis revealed that Lone Wolf attacks resulted in significantly different casualty distributions when compared to both Small Cells ($p < 0.001$) and Large Groups ($p < 0.001$). Notably, there was no statistically significant difference in fatalities between Small Cells and Large Groups ($p = 0.53$). This suggests a critical threshold effect in terrorist operations: while independent actors exhibit a distinct lethality profile, the introduction of even a small network (2-5 perpetrators) shifts the casualty distribution to match that of large-scale group attacks. Consequently, casualty metrics serve as a strong discriminator for imputing independent actors, but offer diminishing returns when attempting to differentiate the size of multi-perpetrator networks.

When examining the effect that tactics and group composition has on casualties in terror attacks, I was concerned that the extreme overdispersion of casualties would result in a standard regression model underestimating the standard errors, leading to false significance. This is a particular risk when considering the imputed dataset, which may already have been

overrepresenting the number of individual perpetrators. As a result, I implemented a Zero-inflated negative binomial (ZINB) model to examine the relationship between nkill and the other predictors. The ZINB model separates the structurally missing (data not determined by GTD) data from the attack data where the attack was a failure. However, upon production of the model, its predictors bias matched the Negative Binomial result. The model slightly under-predicts single fatalities at the expense of an over-prediction of all other values, but overall the model performs well.

Figure 4: Predictors of Independent Actor (Lone Wolf) Terrorism

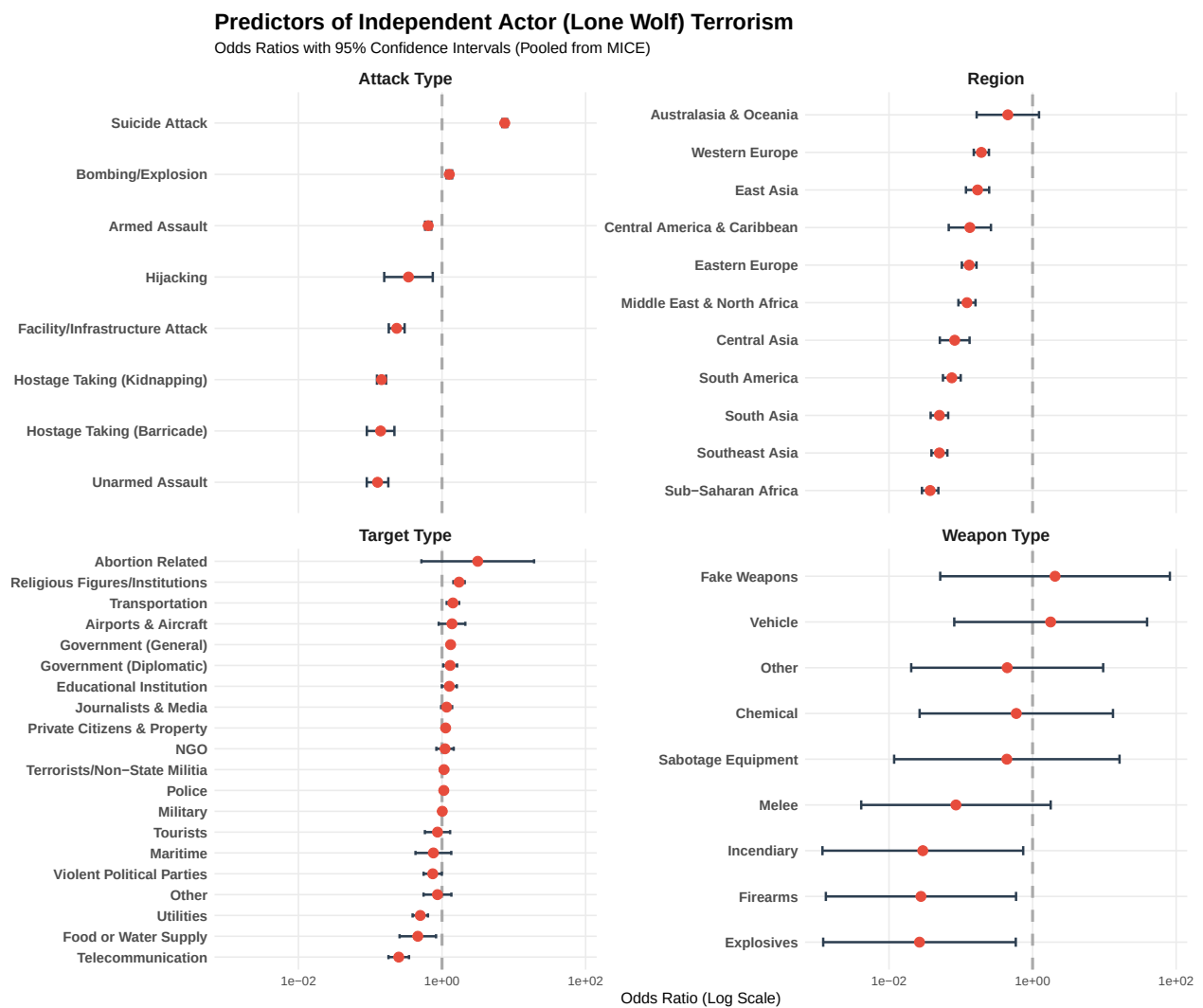


Figure 5: Rootogram comparison demonstrating model fit. The Zero-Inflated model and the Standard Negative Binomial exhibit similar performance

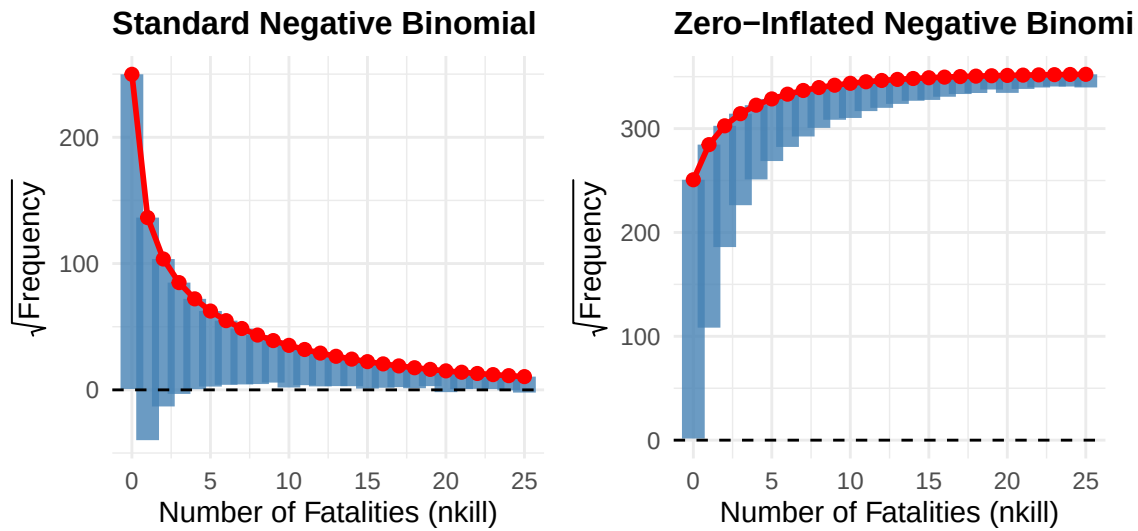
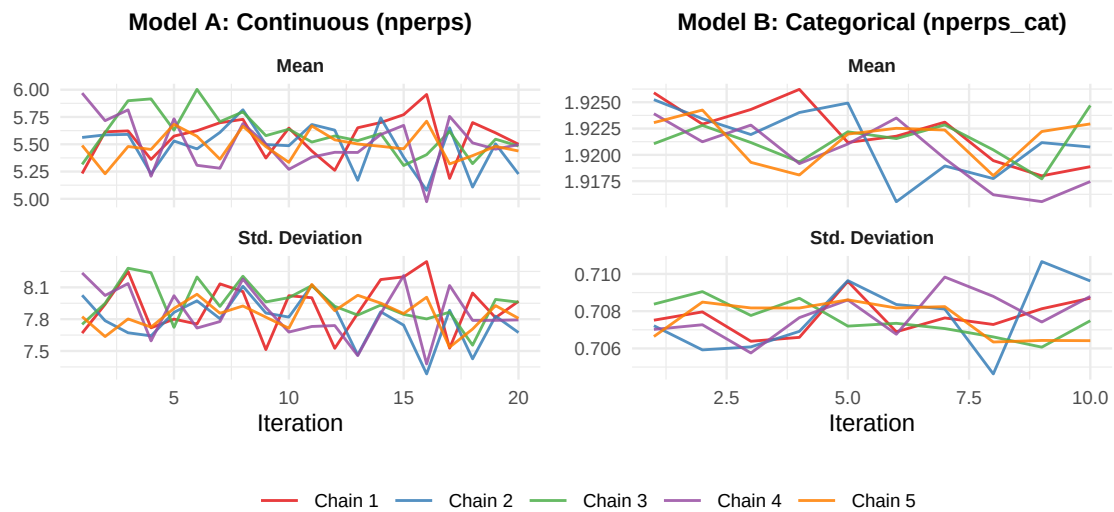


Figure 6: Convergence Trace Plot comparison of MICE imputation models



Call:

```
glm(formula = nperps_missing ~ attacktype1 + weaptype1_cat +
    targtype1 + region + iyear + suicide + nkill + nwound, family = binomial,
    data = gtd_clean)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	17.340999	4.083298	4.247	2.17e-05	***
attacktype1	0.054078	0.007005	7.720	1.17e-14	***
weaptype1_catExplosives	1.497798	0.022435	66.762	< 2e-16	***
weaptype1_catIncendiary	0.262423	0.044834	5.853	4.82e-09	***
weaptype1_catMelee	-1.068801	0.045196	-23.648	< 2e-16	***
weaptype1_catRare/Other	-0.759648	0.094505	-8.038	9.12e-16	***
targtype1	0.017023	0.001582	10.761	< 2e-16	***
region	0.177920	0.003891	45.731	< 2e-16	***
iyear	-0.008821	0.002029	-4.348	1.37e-05	***
suicide	-24.146364	43.772035	-0.552	0.5812	
nkill	0.022236	0.003093	7.190	6.50e-13	***
nwound	-0.002244	0.001051	-2.136	0.0327	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 106436 on 110591 degrees of freedom

Residual deviance: 73710 on 110580 degrees of freedom

(27696 observations deleted due to missingness)

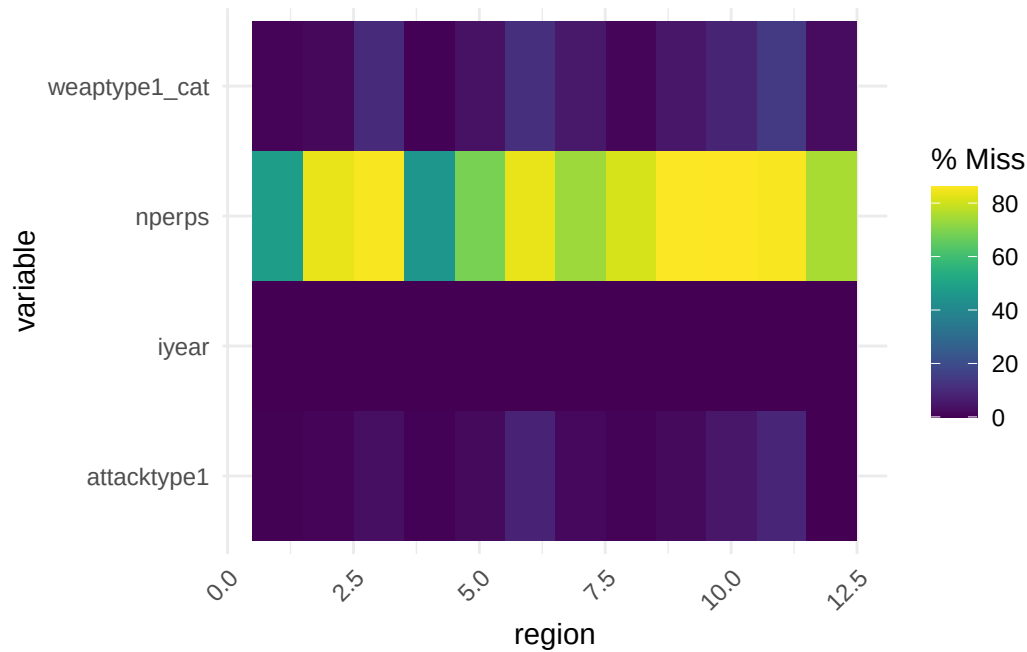
AIC: 73734

Number of Fisher Scoring iterations: 16

fitting null model for pseudo-r2

McFadden

0.3074736



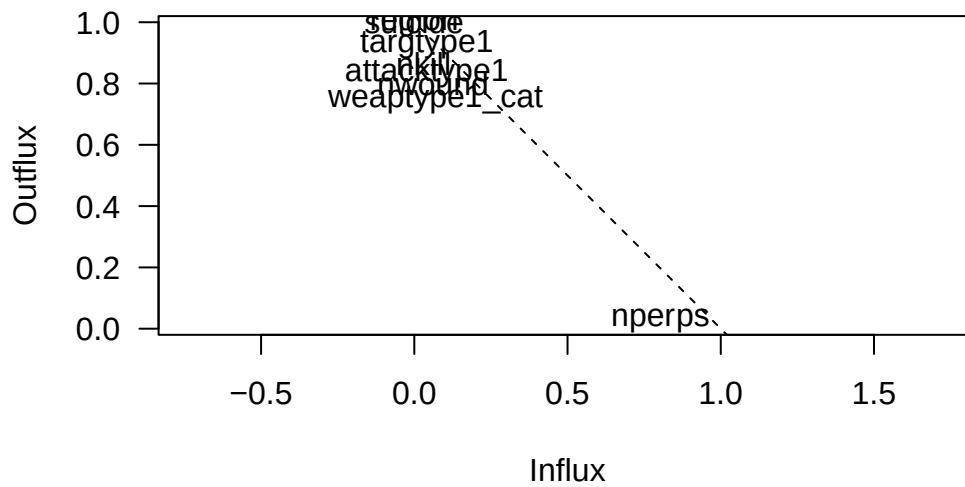
A tibble: 1 x 4

	statistic	df	p.value	missing.patterns
	<dbl>	<dbl>	<dbl>	<int>
1	55351.	126	0	29

```

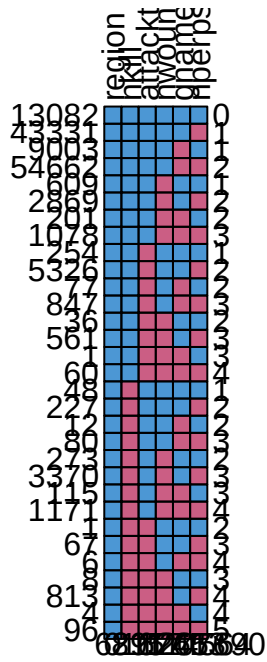
x pattern for gtd_clean %>% dplyr::select(nperps, nkill, nwoi
ux-outflux pattern for   weaptype1_cat, targtype1, region, s

```



PFC	MSE	PFC	MSE	MSE	MSE
0.2954517	1.0595971	0.0872457	31.7626277	0.0000000	0.0000000
MSE	MSE	MSE	MSE		
150.4521868	6826.4635086	0.0000000	0.0000000		

	delta	estimate	conf.low	conf.high
44...1	0.0	6.2481763	5.76072467	6.7768744
44...2	0.5	1.0000000	0.00000000	Inf
44...3	1.0	0.1507358	0.05128243	0.4430619
44...4	1.5	1.0000000	0.00000000	Inf
44...5	2.0	1.0000000	0.00000000	Inf



	region	nkill	attacktype1	nwound	gname	nperps	
13082	1	1	1	1	1	1	0
43331	1	1	1	1	1	0	1
9003	1	1	1	1	0	1	1
54662	1	1	1	1	0	0	2
609	1	1	1	0	1	1	1
2869	1	1	1	0	1	0	2
201	1	1	1	0	0	1	2
1078	1	1	1	0	0	0	3
254	1	1	0	1	1	1	1
5326	1	1	0	1	1	0	2
77	1	1	0	1	0	1	2
847	1	1	0	1	0	0	3
36	1	1	0	0	1	1	2

561	1	1	0	0	1	0	3
1	1	1	0	0	0	1	3
60	1	1	0	0	0	0	4
48	1	0	1	1	1	1	1
227	1	0	1	1	1	0	2
12	1	0	1	1	0	1	2
80	1	0	1	1	0	0	3
273	1	0	1	0	1	1	2
3370	1	0	1	0	1	0	3
115	1	0	1	0	0	1	3
1171	1	0	1	0	0	0	4
1	1	0	0	1	1	1	2
67	1	0	0	1	1	0	3
6	1	0	0	1	0	0	4
8	1	0	0	0	1	1	3
813	1	0	0	0	1	0	4
4	1	0	0	0	0	1	4
96	1	0	0	0	0	0	5
0 6291 8157 11265 67413 114564 207690							

Growing trees.. Progress: 39%. Estimated remaining time: 47 seconds.

Growing trees.. Progress: 80%. Estimated remaining time: 15 seconds.

suicide	weaptype1_cat	attacktype1	region	nkill
0.11059787	0.06607516	0.06129727	0.05588338	0.05020971

```

      nwound      targtype1      iyear_sq      iyear
0.04614369    0.04382867    0.03879369    0.03858311

# A tibble: 2 x 6
  nperps_obs mean_nkill pct_suicide pct_bombing pct_gname_known      n
  <lgl>      <dbl>      <dbl>      <dbl>      <dbl> <int>
1 FALSE          2.11         0         0.543         0.494 114564
2 TRUE           4.18        0.306        0.450         0.603  23724

Likelihood ratio test

Model 1: is.na(nperps) ~ attacktype1 + weaptype1_cat + targtype1 + region +
      iyear + suicide + nkill + nwound
Model 2: is.na(nperps) ~ attacktype1 + weaptype1_cat + targtype1 + region +
      iyear + suicide + nkill + nwound + doc_quality

#Df LogLik Df  Chisq Pr(>Chisq)
1  12 -36855
2  13 -36709  1 291.57 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

3.2 Future work and next steps

The results and efficacy of the model produced by MICE indicates the potential for a comparison analysis against the original data. Next steps will involve incorporating those comparisons as well as performing an analysis of changes over time in the tactics of individual attackers, seeking to find

a causal effect for the increase in lone wolf attacks and their increased ratio of fatalities in recent years.

References

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